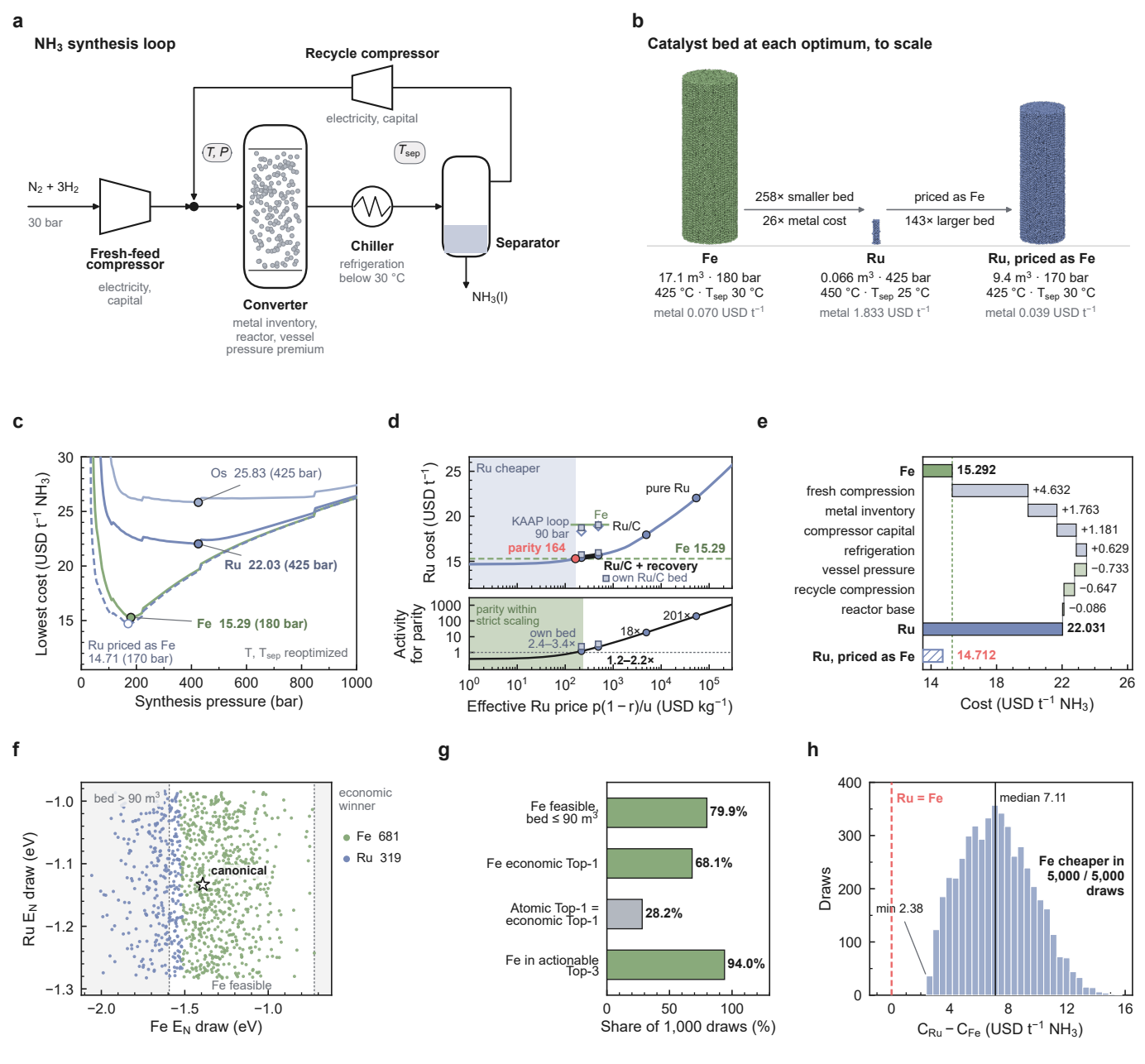

a
NH3 synthesis loop
Recycle compressor
electricity, capital
T, P
Tsep
N2 + 3H2
30 bar
Fresh-feed compressor
electricity, capital
Converter
metal inventory, reactor, vessel pressure premium
Chiller
refrigeration below 30 °C
Separator
NH3(l)
b
Catalyst bed at each optimum, to scale
258× smaller bed
26× metal cost
priced as Fe
143× larger bed
Fe
17.1 m3 · 180 bar
425 °C · Tsep 30 °C
metal 0.070 USD t−1
Ru
0.066 m3 · 425 bar
450 °C · Tsep 25 °C
metal 1.833 USD t−1
Ru, priced as Fe
9.4 m3 · 170 bar
425 °C · Tsep 30 °C
metal 0.039 USD t−1
c
Lowest cost (USD t−1 NH3)
Os 25.83 (425 bar)
Ru 22.03 (425 bar)
Fe 15.29 (180 bar)
Ru priced as Fe
14.71 (170 bar)
T, Tsep reoptimized
Synthesis pressure (bar)
d
Ru cheaper
pure Ru
KAAP loop
90 bar
Fe
Ru/C
parity 164
Fe 15.29
Ru/C + recovery
own Ru/C bed
Ru cost (USD t−1)
parity within strict scaling
own bed
2.4–3.4×
18×
201×
1.2–2.2×
Activity for parity
Effective Ru price p(1 − r)/u (USD kg−1)
e
Fe
15.292
fresh compression
+4.632
metal inventory
+1.763
compressor capital
+1.181
refrigeration
+0.629
vessel pressure
−0.733
recycle compression
−0.647
reactor base
−0.086
Ru
22.031
Ru, priced as Fe
14.712
Cost (USD t−1 NH3)
f
bed > 90 m3
economic winner
Fe 681
Ru 319
canonical
Fe feasible
Ru EN draw (eV)
Fe EN draw (eV)
g
Fe feasible, bed ≤ 90 m3
79.9%
Fe economic Top-1
68.1%
Atomic Top-1 = economic Top-1
28.2%
Fe in actionable Top-3
94.0%
Share of 1,000 draws (%)
h
Ru = Fe
median 7.11
Fe cheaper in 5,000 / 5,000 draws
min 2.38
Draws
CRu − CFe (USD t−1 NH3)